

Impact of Academic and Lifestyle Factors on Student Final Mathematics Grades

STAT 31631 – Statistical Modelling

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Table of Contents

Section	Sub-section
1. Introduction	
	1.1 Background of the Study
	1.2 Problem Statement
	1.3 Objectives
	1.4 Significance of the Study
2. Literature Review	
	2.1 Review of Previous Studies
	2.2 Research Gaps
	2.3 Theoretical and Conceptual Understanding
3. Methodology	
	3.1 Research Design
	3.2 Data Sources and Data Collection Methods
	3.3 Variables Used in the Study
	3.4 Sampling Method
	3.5 Statistical Techniques and Software Used
4. Descriptive Analysis	
	4.1 Sample Overview
	4.2 Summary Statistics
	4.3 Summary of Response Variable (G3)
	4.4 Distribution of G3

Section	Sub-section
	4.5 Categorical Variable Frequencies
	4.6 G3 by Categorical Group
	4.7 Frequency Bar Charts
	4.8 Numeric Predictors vs G3
	4.9 Correlation Matrix
	4.10 Pairwise Relationships
5. Results and Discussion	
	5.1 Present Analytical Results
	5.1.1 Variable Selection Procedure
	5.1.2 Response Variable Transformation to Address Model Assumption Violations
	5.1.2.1 Square-Root Transformation
	5.1.2.2 Cube-Root Transformation
	5.1.2.3 Logarithmic Transformation
	5.1.2.4 Box-Cox Transformation
	5.2 Interpret Findings
	5.2.1 Original Fitted Model – Diagnostic Results and Assumption Violations
	5.2.2 Square-Root Transformation
	5.2.3 Cube-Root Transformation
	5.2.4 Logarithmic Transformation
	5.2.5 Box–Cox Transformation
	5.2.6 Overall Interpretation of the Transformation Results
	5.3 Compare Results with Previous Studies

Section	Sub-section
	5.4 Discuss Implications of the Findings
6. Conclusion and Recommendations	
	6.1 Summary of Major Findings
	6.2 Recommendations
	6.3 Suggested Areas for Future Research
7. References	

List of Tables

Table No.	Title
Table 1	Summary of Previous Studies and Research Gaps
Table 2	Summary Statistics of Numerical Variables
Table 3	Summary Statistics of the Response Variable (G3)
Table 4	Frequencies of Categorical Variables
Table 5	Shapiro–Wilk Normality Test for the Original Fitted Model
Table 6	Breusch–Pagan Test for the Original Fitted Model
Table 7	Durbin–Watson Test for the Original Fitted Model
Table 8	Diagnostic Test Results for the Square-Root Transformed Model
Table 9	Diagnostic Test Results for the Cube-Root Transformed Model
Table 10	Diagnostic Test Results for the Logarithmic Transformed Model
Table 11	Diagnostic Test Results for the Box–Cox Transformed Model

List of Figures

Figure No.	Title
Figure 1	Distribution of Final Grade (G3)
Figure 2	Boxplot of Final Grade (G3)
Figure 3	G3 by Categorical Groups
Figure 4	Frequency Bar Charts of Categorical Variables
Figure 5	Numerical Predictors vs. G3
Figure 6	Correlation Matrix of Numerical Variables
Figure 7	Pairwise Relationships among Selected Numerical Variables
Figure 8	Forward, Backward, and Stepwise Variable Selection Results
Figure 9	Diagnostic Residual Plots for the Original Fitted Model
Figure 10	Fitted Square-Root Transformed Model
Figure 11	Diagnostic Residual Plots for the Square-Root Transformed Model
Figure 12	Fitted Cube-Root Transformed Model
Figure 13	Diagnostic Residual Plots for the Cube-Root Transformed Model
Figure 14	Fitted Logarithmic Transformed Model
Figure 15	Diagnostic Residual Plots for the Logarithmic Transformed Model
Figure 16	Fitted Box–Cox Transformed Model
Figure 17	Diagnostic Residual Plots for the Box–Cox Transformed Model

1 Introduction

1.1 Background of the Study

How well students do in school depends on many things, not just how smart they are. Things like family background, social life, school environment, and past academic performance can all affect grades. This project looks at data from students in two Portuguese schools, first collected by Cortez and Silva (2008) using school records and student surveys. We are using the Mathematics dataset. It includes information about each student's background (age, sex, where they live), family (parents' education and jobs), lifestyle (study time, free time, drinking habits, absences from school), and their grades from three points in the year: G1 (first term), G2 (second term), and G3 (final grade).

1.2 Problem Statement

A student's final grade (G3) is shaped by many different factors, and it's useful to understand exactly which ones matter most and how strongly. Since G1 and G2 are recorded earlier in the school year, they are a natural and highly informative starting point for predicting G3 — but it is not yet clear which additional background, family, and lifestyle factors continue to matter once prior grades are accounted for. This project addresses the problem of building an accurate and interpretable regression model that predicts the final Mathematics grade (G3) using prior-period grades (G1, G2) together with demographic, social, and school-related information.

1.3 Objectives

The main objective of this project is to build a strong regression model to predict students' final Mathematics grade (G3), using G1, G2, and other background, family, and school-related variables as predictors. The specific goals are:

- Explore and summarize the data using statistics and graphs.
- Find out which factors (including prior grades) have the biggest effect on the final grade.
- Try different regression models (forward, backward, stepwise selection) and choose the best one.
- Check how well the chosen model performs using proper diagnostic tests.
- Explain what the results mean and give simple recommendations.

1.4 Significance of the Study

This project helps identify which factors, beyond prior academic performance, continue to matter for a student's final grade. Since G1 and G2 explain most of the variation in G3, including them allows the model to also detect smaller but still meaningful effects (such as failures, school, or travel time) that remain significant even after accounting for prior

performance. This is useful for teachers and schools who already have access to G1/G2 and want the most accurate possible prediction of a student's final outcome, along with insight into which other factors still deserve attention.

2. Literature Review

2.1 Review of Previous Studies

Research	Methodology	Key Conclusion	Research Gap in Relation to the Present Study
Zubair et al. (2024) – Study Hours & Academic Performance	Survey of 209 students; students grouped according to weekly study hours.	Students studying approximately 4–6 hours per week achieved better academic performance, indicating an association between study time and grades.	Primarily focuses on study hours, whereas student performance may be related to a broader combination of academic, demographic, family, social and school-related characteristics.
Gong et al. (2025) – Social Media & Academic Performance	Structural equation modelling involving 3,716 undergraduate students.	High-frequency social-media use was associated with poorer academic performance through social anxiety and FOMO-related pathways.	Focuses mainly on social-media and psychological factors rather than modelling final academic performance using a broad set of student characteristics.
Faught et al. (2017) – Lifestyle Behaviours & Academic Performance	Prospective survey of 4,253 Grade 5 students; standardized academic assessments.	Lifestyle behaviours such as diet, sleep and physical activity were associated with academic performance.	Concentrates primarily on lifestyle-related factors and a younger student population, leaving scope to examine academic performance using a wider combination of student-related characteristics.
Oughdir et al. (2020) – Prediction of Student Performance Using MLR	Multiple linear regression with k-fold cross-validation.	Multiple linear regression was used to predict student performance, with cross-validation supporting the reliability of the predictive approach.	Demonstrates the usefulness of MLR, but there remains scope to determine a suitable regression model for final grade using the combination of predictors available in the present dataset.
Riaz et al. (2023) – Screen Time, Social Media & Academic Performance	Quantitative survey of 200 university students; SPSS-based statistical analysis.	Social-media and entertainment-related behaviour was associated with students' academic success.	Mainly examines screen time and social-media behaviour among university students rather than simultaneously considering academic, demographic, family, social and school-related factors.

Table_1

2.2 Research Gaps

Previous studies have examined academic performance by focusing on specific factors such as study hours, social media usage, lifestyle behaviours, screen time, and entertainment-related activities. Although these studies provide useful evidence regarding individual factors associated with academic achievement, many focus on a particular group of factors rather than considering a wider range of student characteristics simultaneously.

Furthermore, previous research has demonstrated the usefulness of multiple linear regression and other statistical techniques for predicting student performance. However, there remains scope to investigate how a broader combination of academic, demographic, family, social, and school-related characteristics can be used together to model students' final academic performance.

Therefore, the present study addresses this gap by using the UCI Student Performance dataset and considering G3 (final grade) as the response variable. The study examines multiple student-related characteristics simultaneously with the aim of identifying a suitable regression model for final academic performance.

2.3 Theoretical and Conceptual Understanding

This project is guided by the general idea that academic performance is shaped by a combination of a student's prior academic trajectory and their personal, family, and environmental circumstances. Prior grades (G1 and G2) are treated as a strong summary indicator of a student's ongoing academic trajectory, reflecting the cumulative influence of ability, effort, and circumstances up to that point in the year. At the same time, the project draws on the Family Investment Model, which suggests that families with more resources (income, education, time) can support their children's education more effectively, leading to better academic outcomes, as well as general education research showing that behavioural factors such as study time, class failures, and school environment can directly affect how well a student performs, independently of their prior grades.

In terms of statistical modelling, this project follows the same conceptual approach used in prior regression-based studies on this dataset: since G3 is a continuous numeric variable (grades from 0–20), multiple linear regression is a suitable technique to model the relationship between G3 and its predictors. Including G1 and G2 as predictors allows the model to first account for the strong influence of prior academic performance, so that the estimated effects of the remaining background and lifestyle variables can be interpreted as the additional impact of each factor beyond what is already explained by a student's earlier grades.

3. Methodology

3.1 Research Design

This study follows a quantitative, explanatory research design based on secondary data. The aim is to model the relationship between a student's final Mathematics grade (G3) and a set of prior academic grades, demographic, family, social, and school-related predictor variables, using multiple linear regression. Since the goal is prediction and explanation rather than experimentation, no treatment or intervention is applied to the students; the analysis works entirely with data that was already collected and recorded.

The following assumptions of multiple linear regression are considered and need to be satisfied to fit an appropriate and reliable model:

- Linearity between the predictors and the response variable (G3).
- Independence of observations.
- Constant variance of errors (homoscedasticity).
- Normality of residuals.
- No serious multicollinearity among the predictor variables.
- No influential outliers that could strongly affect the fitted model.

3.2 Data Sources and Data Collection Methods

The data used in this study was originally collected by Cortez and Silva (2008) from two secondary schools in Portugal Gabriel Pereira (GP) and Mousinho da Silveira (MS) and is publicly available through the UCI Machine Learning Repository. The dataset was gathered using two main methods: school reports, which provided official academic records (such as grades, absences, and school-related information), and student questionnaires, which collected demographic, family, and lifestyle information (such as parents' education, study time, and alcohol consumption). This project uses the Mathematics (mat) subset of the dataset.

3.3 Variables Used in the Study

The dataset contains 33 variables in total. The response (dependent) variable for this study is:

- G3 – the student's final Mathematics grade, measured on a numeric scale from 0 to 20.

The predictor (independent) variables considered fall into four broad categories:

- Prior academic performance variables: G1 (first-period grade) and G2 (second-period grade), both measured on the same 0–20 scale as G3

- Demographic variables: school, sex, age, address (urban/rural), family size, parents' cohabitation status.
- Family and social background variables: mother's and father's education level, mother's and father's job, guardian, family relationship quality, going out with friends, weekend/workday alcohol consumption, and current health status.
- School related and lifestyle variables: study time, number of past class failures, reason for choosing the school, extra educational support, family educational support, extra paid classes, participation in extracurricular activities, internet access at home, romantic relationship status, travel time to school, free time after school, and number of school absences.

Note that G1 and G2 are deliberately included as predictors in this study. Since G3 is strongly correlated with a student's grades earlier in the year, including G1 and G2 allows the model to achieve substantially higher predictive accuracy, while also making it possible to identify which additional demographic, family, and school-related factors remain statistically significant even after prior academic performance is accounted for.

3.4 Sampling Method

No additional sampling was carried out for this study, as the data was used as originally collected by Cortez and Silva (2008). The original dataset was gathered through a convenience sampling approach, where students from the two participating schools who completed both the school-report data collection and the questionnaire were included. This project uses the full Mathematics dataset as provided, without drawing a further sub-sample.

3.5 Statistical Techniques and Software Used

Descriptive statistics (means, standard deviations, frequency tables, and visualizations such as histograms, boxplots, and correlation plots) were first used to explore the data and check its main characteristics.

For model building, multiple linear regression was used to model G3 as a function of the selected predictor variables. Since the dataset contains a large number of candidate predictors, three variable selection procedures were applied to identify the most suitable subset of predictors:

- Forward selection – starting with no predictors and adding variables one at a time based on the greatest improvement in model fit, until no further significant improvement is found.
- Backward elimination – starting with all candidate predictors included and removing the least significant variable at each step, until only significant predictors remain.
- Stepwise selection – a combination of forward and backward steps, where variables can be added or removed at each stage, allowing the procedure to reconsider earlier decisions.

Each of the three resulting models was compared using model selection criteria such as Adjusted R^2 , AIC (Akaike Information Criterion), BIC (Bayesian Information Criterion), and residual standard error, along with diagnostic checks (residual plots, normality of residuals, multicollinearity via VIF, and homoscedasticity) to confirm the assumptions of linear regression were reasonably satisfied. The model with the best balance of simplicity, fit, and statistical validity was selected as the final model.

All data cleaning, exploratory analysis, and statistical modelling were carried out using R (with packages such as MASS, car, and leaps commonly used for stepwise selection and regression diagnostics), consistent with the course's use of R for statistical modelling.

4. Descriptive Analysis

4.1 Sample overview

The dataset contains 649 observations and 33 variables.

4.2 Summary statistics

age	Medu	Fedu	traveltime	studytime	failures
Min. :15.00	Min. :0.000	Min. :0.000	Min. :1.000	Min. :1.000	Min. :0.0000
1st Qu.:16.00	1st Qu.:2.000	1st Qu.:1.000	1st Qu.:1.000	1st Qu.:1.000	1st Qu.:0.0000
Median :17.00	Median :2.000	Median :2.000	Median :1.000	Median :2.000	Median :0.0000
Mean :16.74	Mean :2.515	Mean :2.307	Mean :1.569	Mean :1.931	Mean :0.2219
3rd Qu.:18.00	3rd Qu.:4.000	3rd Qu.:3.000	3rd Qu.:2.000	3rd Qu.:2.000	3rd Qu.:0.0000
Max. :22.00	Max. :4.000	Max. :4.000	Max. :4.000	Max. :4.000	Max. :3.0000
famrel	freetime	goout	Dalc	Walc	health
Min. :1.000	Min. :1.00	Min. :1.000	Min. :1.000	Min. :1.00	Min. :1.000
1st Qu.:4.000	1st Qu.:3.00	1st Qu.:2.000	1st Qu.:1.000	1st Qu.:1.00	1st Qu.:2.000
Median :4.000	Median :3.00	Median :3.000	Median :1.000	Median :2.00	Median :4.000
Mean :3.931	Mean :3.18	Mean :3.185	Mean :1.502	Mean :2.28	Mean :3.536
3rd Qu.:5.000	3rd Qu.:4.00	3rd Qu.:4.000	3rd Qu.:2.000	3rd Qu.:3.00	3rd Qu.:5.000
Max. :5.000	Max. :5.00	Max. :5.000	Max. :5.000	Max. :5.00	Max. :5.000
absences	G1	G2	G3		
Min. : 0.000	Min. : 0.0	Min. : 0.00	Min. : 0.00		
1st Qu.: 0.000	1st Qu.:10.0	1st Qu.:10.00	1st Qu.:10.00		
Median : 2.000	Median :11.0	Median :11.00	Median :12.00		
Mean : 3.659	Mean :11.4	Mean :11.57	Mean :11.91		
3rd Qu.: 6.000	3rd Qu.:13.0	3rd Qu.:13.00	3rd Qu.:14.00		
Max. :32.000	Max. :19.0	Max. :19.00	Max. :19.00		

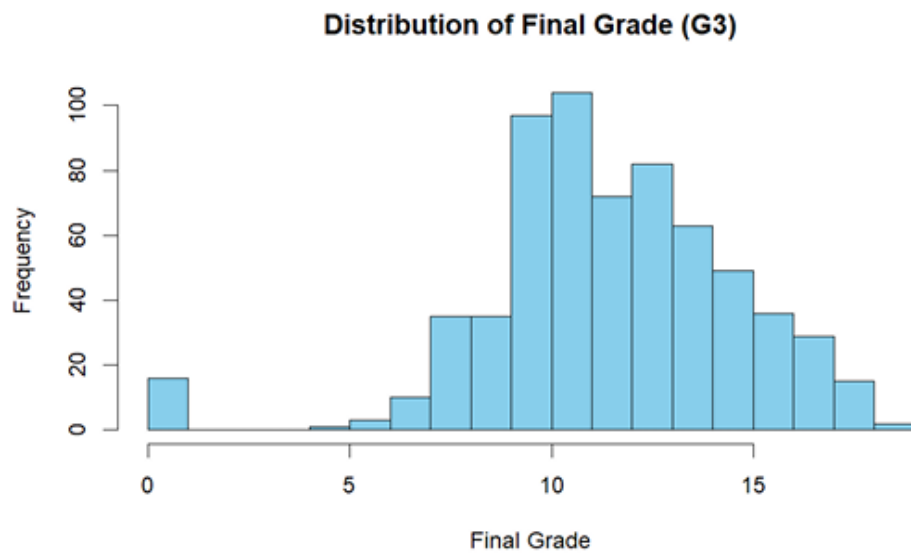
Table_2

4.3 Summary of response variable (G3)

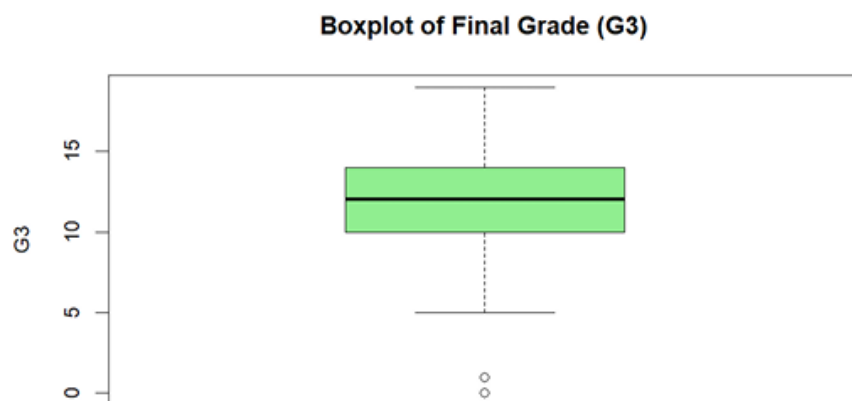
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.00	10.00	12.00	11.91	14.00	19.00
[1]	3.230656					
[1]	10.43714					

Table_3

4.4 Distribution of G3



Figure_1



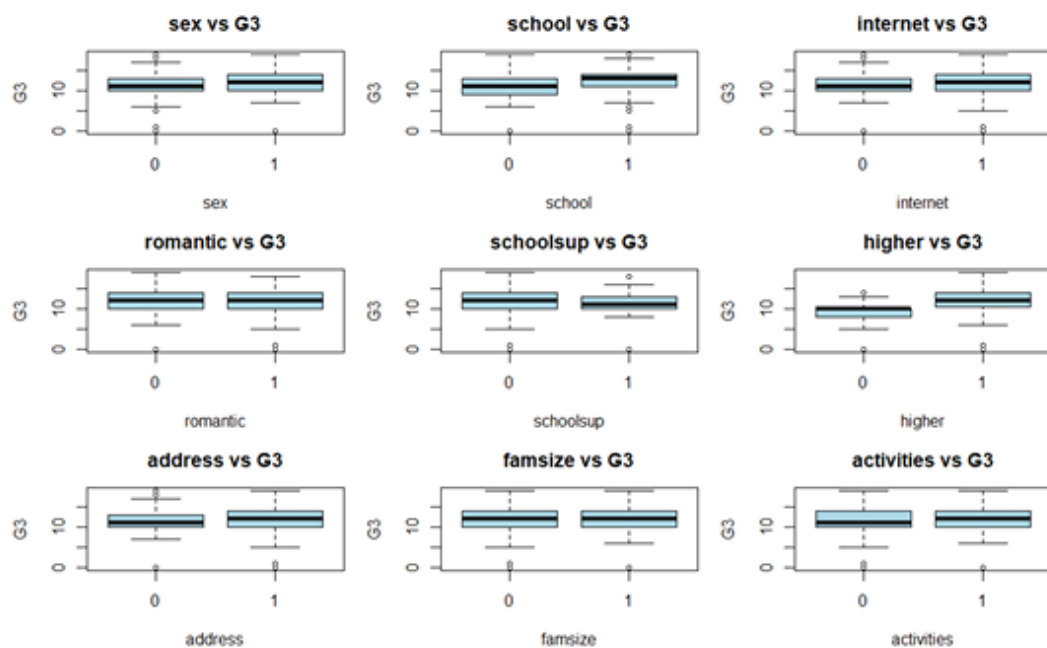
Figure_2

4.5 Categorical variable frequencies

	F	M			
	383	266			
	GP	MS			
	423	226			
	R	U			
	197	452			
at_home	health	other	services	teacher	
135	48	258	136	72	
at_home	health	other	services	teacher	
42	23	367	181	36	

Table_4

4.6 G3 by categorical group



Figure_3

4.7 Frequency bar charts

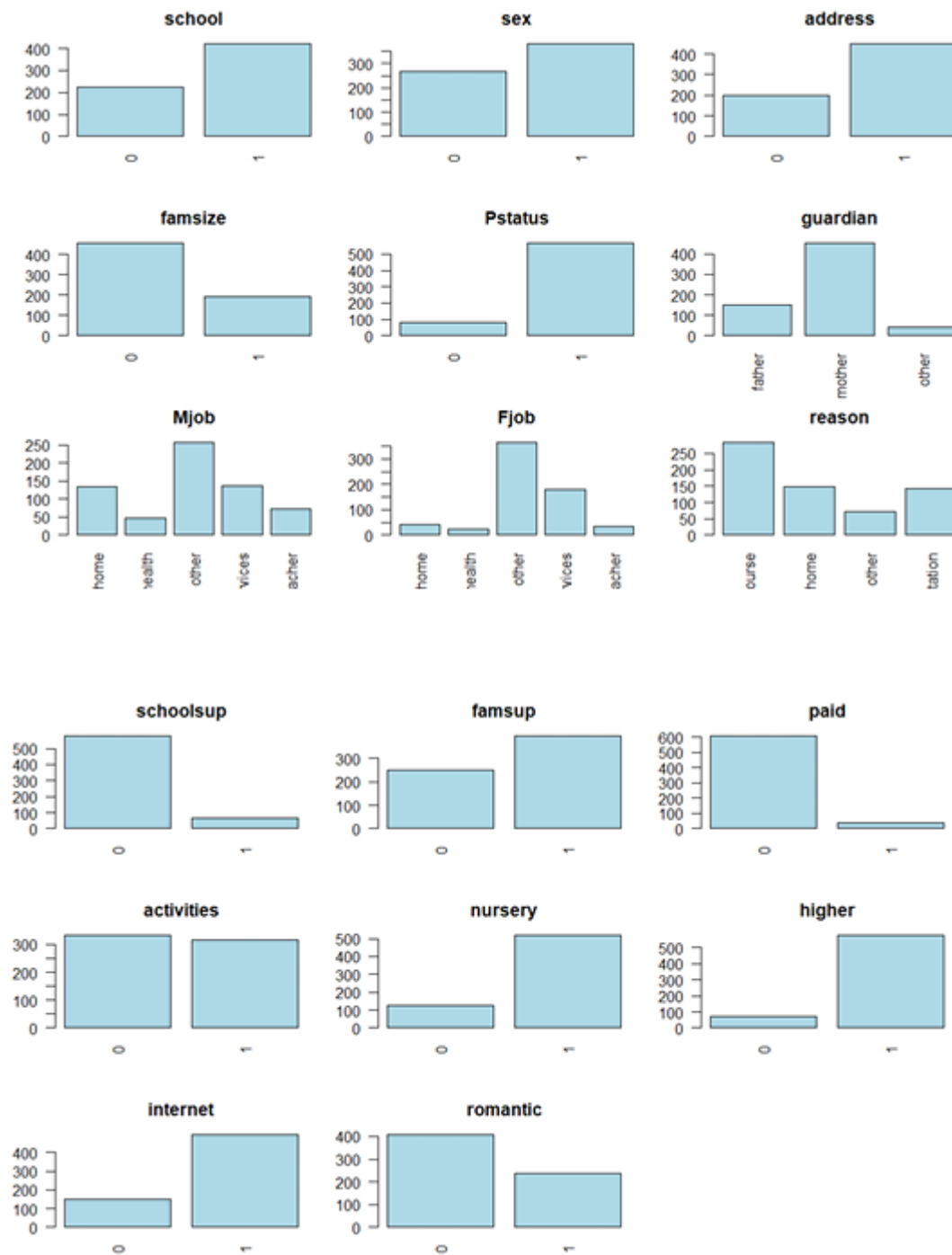
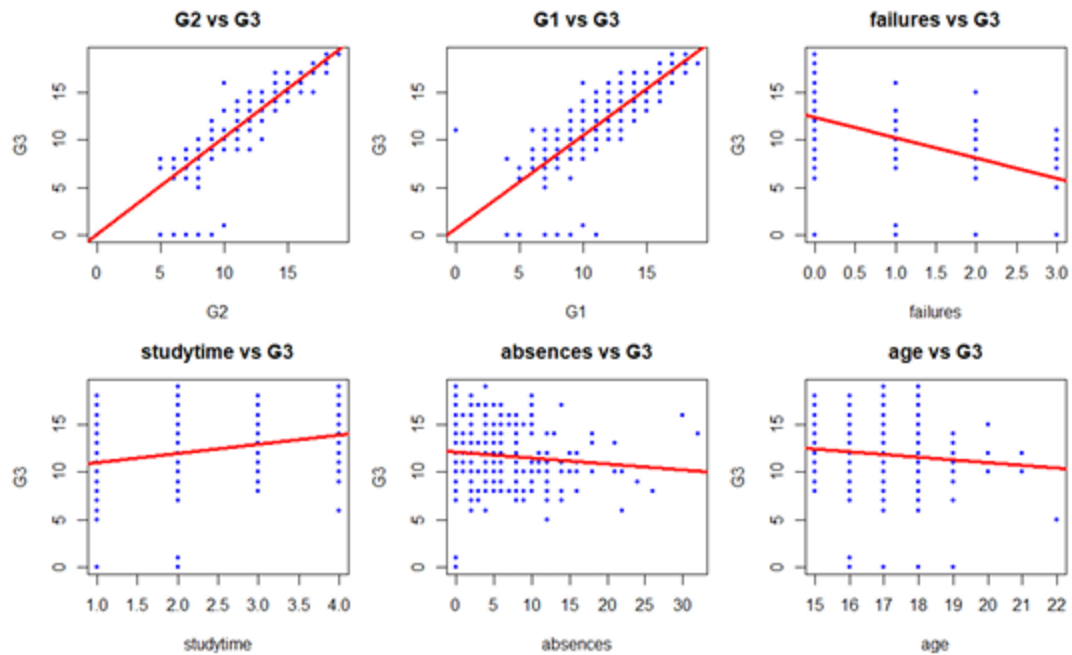


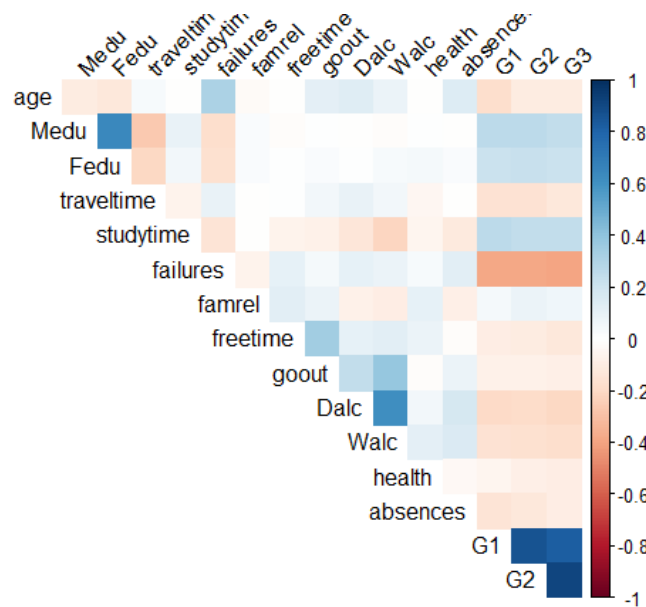
Figure 4

4.8 Numeric predictors vs G3



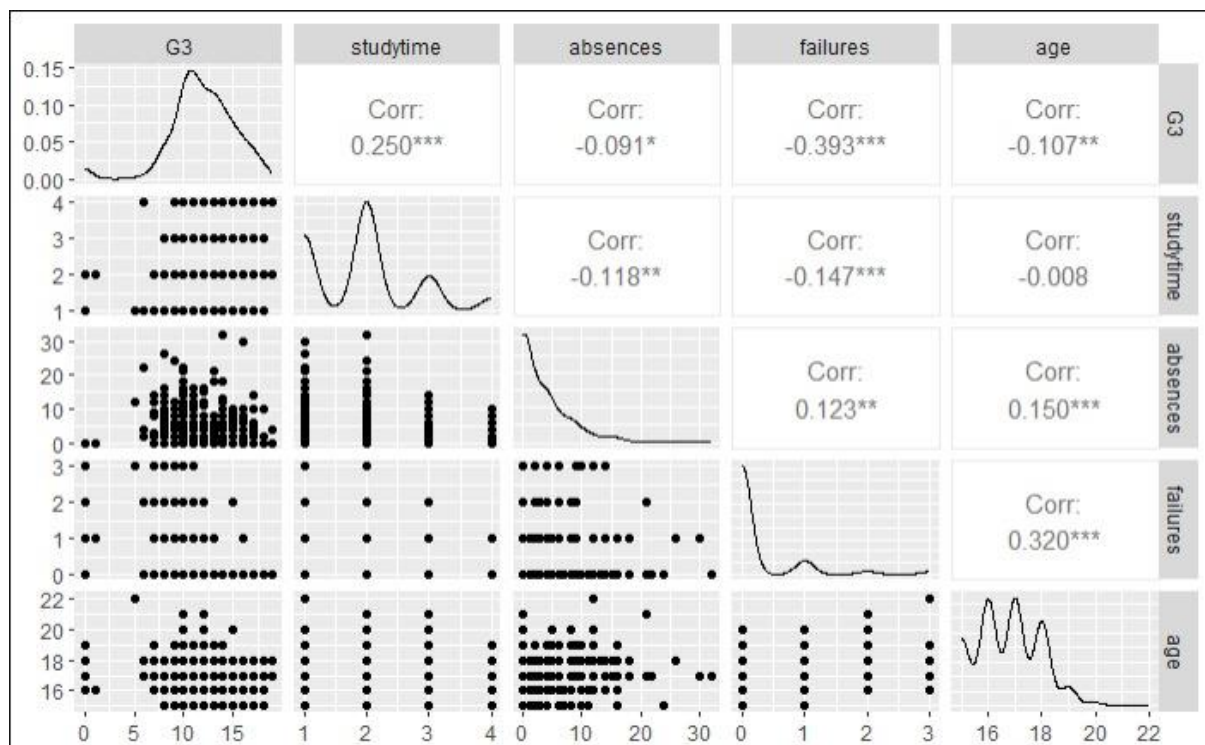
Figure_5

4.9 Correlation matrix



Figure_6

4.10 Pairwise relationships



Figure_7

The descriptive analysis suggests failures, study time, school, sex, and parental education are the strongest candidates for significant predictors in your regression model, while address and job categories appear to have weaker or less consistent relationships with G3 useful groundwork to compare against what your forward/backward/stepwise selection actually picks in the Results section.

5 Results and Discussion

5.1 Present Analytical Results

5.1.1 Variable Selection Procedure

Three variable selection procedures (Forward, Backward, and Stepwise) were applied to identify the best regression model for predicting students' final Mathematics grade. All three methods converged on the same final model, with variables listed in different order but identical coefficients, confirming this model is a stable, well-supported choice regardless of selection direction.

```
Call:
lm(formula = G3 ~ school + sex + reason + traveltime + failures +
    health + absences + G1 + G2, data = math_data)

Residuals:
    Min       1Q   Median       3Q      Max
-9.0833 -0.5178 -0.0053  0.6398  5.2097

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.44063    0.34169   1.290 0.197678
schoolMS      -0.22981    0.11621  -1.977 0.048419 *
sexM          -0.20022    0.10191  -1.965 0.049894 *
reasonhome    -0.09222    0.13010  -0.709 0.478659
reasonother   -0.44994    0.16627  -2.706 0.006990 **
reasonreputation -0.16537    0.13290  -1.244 0.213816
traveltime    -0.11228    0.06839  -1.642 0.101138
failures      -0.24049    0.09074  -2.650 0.008244 **
health        -0.05394    0.03469  -1.555 0.120451
absences      -0.01623    0.01100   1.476 0.140522
G1            0.13706    0.03615   3.792 0.000164 ***
G2            0.87996    0.03379  26.042 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.242 on 637 degrees of freedom
Multiple R-squared:  0.8547,    Adjusted R-squared:  0.8522
F-statistic: 340.7 on 11 and 637 DF,  p-value: < 2.2e-16
```

Backward Elimination Method

```
Call:
lm(formula = G3 ~ G2 + G1 + failures + reason + absences + sex +
    school + traveltime + health, data = math_data)

Residuals:
    Min       1Q   Median       3Q      Max
-9.0833 -0.5178 -0.0053  0.6398  5.2097

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.44063    0.34169   1.290 0.197678
G2            0.87996    0.03379  26.042 < 2e-16 ***
G1            0.13706    0.03615   3.792 0.000164 ***
failures      -0.24049    0.09074  -2.650 0.008244 **
reasonhome    -0.09222    0.13010  -0.709 0.478659
reasonother   -0.44994    0.16627  -2.706 0.006990 **
reasonreputation -0.16537    0.13290  -1.244 0.213816
absences      -0.01623    0.01100   1.476 0.140522
sexM          -0.20022    0.10191  -1.965 0.049894 *
schoolMS      -0.22981    0.11621  -1.977 0.048419 *
traveltime    -0.11228    0.06839  -1.642 0.101138
health        -0.05394    0.03469  -1.555 0.120451
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.242 on 637 degrees of freedom
Multiple R-squared:  0.8547,    Adjusted R-squared:  0.8522
F-statistic: 340.7 on 11 and 637 DF,  p-value: < 2.2e-16
```

Forward Selection Method

```
Call:
lm(formula = G3 ~ G2 + G1 + failures + reason + absences + sex +
    school + traveltime + health, data = math_data)

Residuals:
    Min       1Q   Median       3Q      Max
-9.0833 -0.5178 -0.0053  0.6398  5.2097

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.44063    0.34169   1.290 0.197678
G2            0.87996    0.03379  26.042 < 2e-16 ***
G1            0.13706    0.03615   3.792 0.000164 ***
failures      -0.24049    0.09074  -2.650 0.008244 **
reasonhome    -0.09222    0.13010  -0.709 0.478659
reasonother   -0.44994    0.16627  -2.706 0.006990 **
reasonreputation -0.16537    0.13290  -1.244 0.213816
absences      -0.01623    0.01100   1.476 0.140522
sexM          -0.20022    0.10191  -1.965 0.049894 *
schoolMS      -0.22981    0.11621  -1.977 0.048419 *
traveltime    -0.11228    0.06839  -1.642 0.101138
health        -0.05394    0.03469  -1.555 0.120451
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.242 on 637 degrees of freedom
Multiple R-squared:  0.8547,    Adjusted R-squared:  0.8522
F-statistic: 340.7 on 11 and 637 DF,  p-value: < 2.2e-16
```

Stepwise Selection Procedure

Figure_8

Fitted model equation:

$$G3 = 0.441 + 0.880(G2) + 0.137(G1) - 0.240(\text{failures}) - 0.092(\text{reasonhome}) - 0.450(\text{reasonother}) - 0.165(\text{reasonreputation}) + 0.016(\text{absences}) - 0.200(\text{sexM}) - 0.230(\text{schoolMS}) + 0.112(\text{traveltime}) - 0.054(\text{health}) + \varepsilon$$

Overall model fit: $R^2 = 0.8547$, Adjusted $R^2 = 0.8522$, Residual Standard Error = 1.242 on 637 degrees of freedom, $F(11, 637) = 340.7$, $p < 2.2e-16$ an extremely strong and highly statistically significant model.

Significance summary:

5 predictors are significant at conventional levels (G2, G1, failures, reasonother, and marginally sexM/schoolMS at $p \approx 0.05$); the remaining variables (reasonhome, reasonreputation, absences, traveltime, health) are not statistically significant in this model.

Formals Statistical Tests for check the assumptions:

- **Shapiro-Wilk Normality Test:**

H0: The residuals are normally distributed.

H1: The residuals are not normally distributed.

```
shapiro-wilk normality test
data:  residuals(forward_model)
W = 0.77153, p-value < 2.2e-16
```

Table_5

- **Breusch-Pagan test for homoscedasticity**

H0: The residuals have constant variance (homoscedasticity).

H1: The residuals do not have constant variance (heteroscedasticity).

```
studentized Breusch-Pagan test
data:  forward_model
BP = 33.64, df = 11, p-value = 0.000414
```

Table_6

- **Durbin-Watson test for independence.**

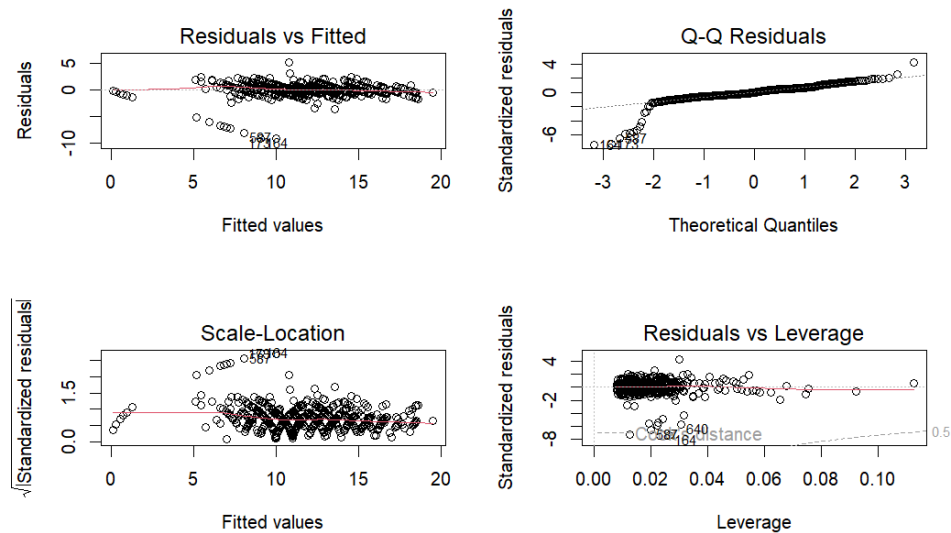
H0: The residuals are independent (no autocorrelation).

H1: The residuals are not independent (autocorrelation is present).

```
Durbin-Watson test
data:  forward_model
DW = 1.8628, p-value = 0.03315
alternative hypothesis: true autocorrelation is greater than 0
```

Table_7

Residual Plots for the fitted model:



Figure_9

5.1.2 Response Variable Transformation to Address Model Assumption Violations

Since the diagnostic analysis indicated violations of the linear regression assumptions, several transformations of the response variable were investigated to determine whether the model assumptions could be improved. The following transformations were considered:

- Square-root transformation: $\sqrt{Y}$
- Cube-root transformation: $\sqrt[3]{Y}$
- Logarithmic transformation: $\log(Y)$
- Box–Cox transformation

After applying each transformation, the regression model was refitted and the model assumptions were reassessed using diagnostic plots, the Shapiro–Wilk test, the Breusch–Pagan test, and the Durbin–Watson test.

5.1.2.1 Square-Root Transformation

Fitted Model:

```
Call:
lm(formula = sqrtG3 ~ G2 + G1 + failures + reason + sex + school,
    data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-2.08665 -0.07276  0.01673  0.12315  0.73372

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.874532   0.059605  31.449 < 2e-16 ***
G2           0.157832   0.007362  21.438 < 2e-16 ***
G1          -0.006347   0.007853  -0.808  0.41925
failures    -0.051409   0.019755  -2.602  0.00947 **
reasonhome  -0.024863   0.028022  -0.887  0.37527
reasonother -0.098249   0.036302  -2.706  0.00698 **
reasonreputation -0.046757  0.028692  -1.630  0.10367
sexM        -0.035735   0.022032  -1.622  0.10530
schoolMS    -0.068358   0.024303  -2.813  0.00506 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2714 on 640 degrees of freedom
Multiple R-squared:  0.7517,    Adjusted R-squared:  0.7486
F-statistic: 242.2 on 8 and 640 DF,  p-value: < 2.2e-16
```

Figure_10

Formal Statistical Tests for check the assumptions:

```
===== SQUARE ROOT TRANSFORMATION DIAGNOSTICS =====

1. Shapiro-Wilk Normality Test:
    Shapiro-Wilk normality test
data:  residuals(model_sqrt)
W = 0.64667, p-value < 2.2e-16

2. Breusch-Pagan Test (Heteroscedasticity):
    studentized Breusch-Pagan test
data:  model_sqrt
BP = 44.698, df = 8, p-value = 4.199e-07

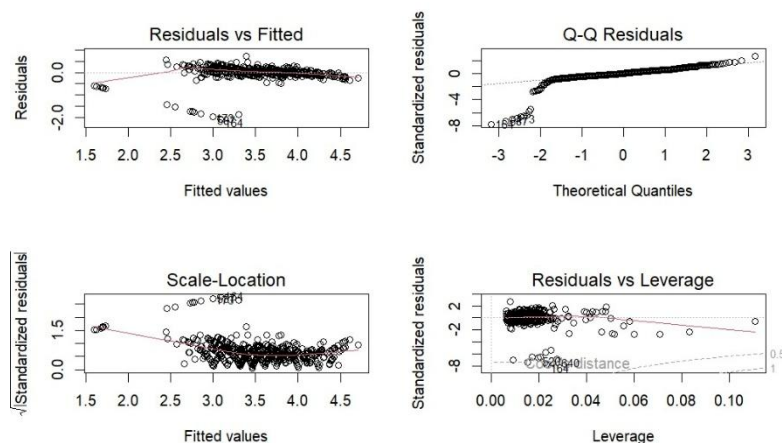
3. VIF (Multicollinearity):
    GVIF Df GVIF^(1/(2*Df))
G2      4.048263 1      2.012030
G1      4.089113 1      2.022156
failures 1.208265 1      1.099211
reason   1.141271 3      1.022268
sex      1.034495 1      1.017101
school   1.181198 1      1.086830

4. Durbin-Watson Test (Autocorrelation):
    Durbin-Watson test
data:  model_sqrt
DW = 1.8874, p-value = 0.06485
alternative hypothesis: true autocorrelation is greater than 0

5. Model Fit:
R-squared: 0.7517
Adjusted R-squared: 0.7486
AIC: 159.91
BIC: 204.66
Residual SE: 0.2714
```

Table_8

Residual Plots for the fitted model:



Figure_11

5.1.2.2 Cube-Root Transformation

Fitted Model:

```
Call:
lm(formula = cbrtg3 ~ G2 + G1 + failures + reason + sex + school,
    data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.09681 -0.03349  0.01131  0.06174  0.32003

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.563362   0.031643  49.407 < 2e-16 ***
G2           0.077135   0.003909  19.735 < 2e-16 ***
G1          -0.008345   0.004169  -2.002  0.04574 *
failures     -0.026080   0.010487  -2.487  0.01314 *
reasonhome   -0.014190   0.014876  -0.954  0.34050
reasonother  -0.050872   0.019272  -2.640  0.00850 **
reasonreputation -0.026548  0.015232  -1.743  0.08183 .
sexM         -0.016802   0.011696  -1.437  0.15133
schoolMS     -0.038252   0.012902  -2.965  0.00314 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1441 on 640 degrees of freedom
Multiple R-squared:  0.6985,    Adjusted R-squared:  0.6948
F-statistic: 185.4 on 8 and 640 DF,  p-value: < 2.2e-16
```

Figure_12

Formal Statistical Tests for check the assumptions:

```
===== CUBE ROOT TRANSFORMATION DIAGNOSTIC TESTS =====

1. Shapiro-Wilk Normality Test:
    Shapiro-wilk normality test
data:  residuals(model_cbrt)
W = 0.61779, p-value < 2.2e-16

2. Breusch-Pagan Test (Heteroscedasticity):
    studentized Breusch-Pagan test
data:  model_cbrt
BP = 52.674, df = 8, p-value = 1.247e-08

3. VIF (Multicollinearity):
      GVIF Df GVIF^(1/(2*Df))
G2      4.048263  1      2.012030
G1      4.089113  1      2.022156
failures 1.208265  1      1.099211
reason   1.141271  3      1.022268
sex      1.034495  1      1.017101
school   1.181198  1      1.086830

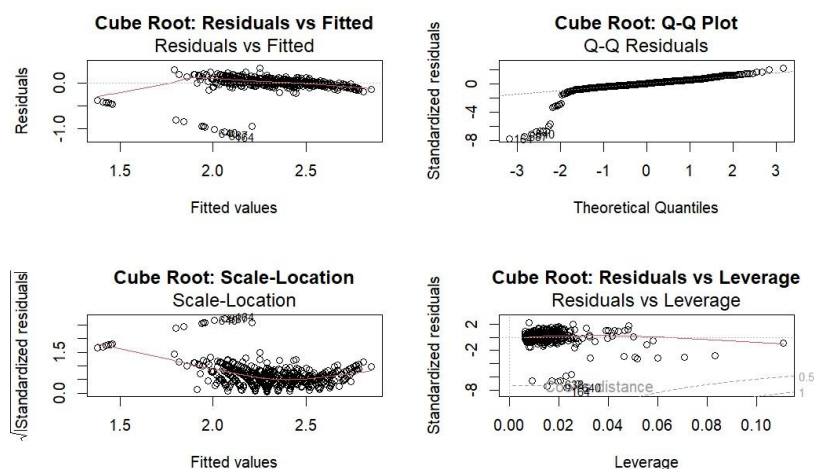
4. Durbin-Watson Test (Autocorrelation):
    Durbin-Watson test
data:  model_cbrt
DW = 1.8836, p-value = 0.05889
alternative hypothesis: true autocorrelation is greater than 0

5. Model Fit Measures:
R-squared: 0.6985
Adjusted R-squared: 0.6948
AIC: -662.01
BIC: -617.26
Residual Standard Error: 0.1441

6. Coefficients:
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.563361607 0.031642805 49.4065429 1.396436e-220
G2           0.077134737 0.003908524 19.7350016 4.464944e-68
G1          -0.008345295 0.004169120 -2.0016921 4.573947e-02
failures     -0.026079830 0.010487415 -2.4867739 1.314451e-02
reasonhome   -0.014190322 0.014876159 -0.9538969 3.404960e-01
reasonother  -0.050872382 0.019271890 -2.6397194 8.499658e-03
reasonreputation -0.026548142 0.015232058 -1.7429123 8.182920e-02
sexM         -0.016802411 0.011696287 -1.4365593 1.513319e-01
schoolMS     -0.038251935 0.012902118 -2.9647794 3.141701e-03
```

Table_9

Residual Plots for the fitted model:



Figure_13

5.1.2.3 Logarithmic Transformation

Fitted Model:

```
Call:
lm(formula = logG3 ~ G2 + G1 + failures + reason + sex + school,
    data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-2.13497 -0.05789  0.03084  0.11780  0.55961

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.419735   0.063549   22.341 < 2e-16 ***
G2           0.131701   0.007850   16.778 < 2e-16 ***
G1          -0.032123   0.008373   -3.837 0.000137 ***
failures    -0.045355   0.021062   -2.153 0.031659 *
reasonhome  -0.031674   0.029876   -1.060 0.289464
reasonother -0.096101   0.038704   -2.483 0.013284 *
reasonreputation -0.057731 0.030591   -1.887 0.059584 .
sexM        -0.026006   0.023490   -1.107 0.268665
schoolMS    -0.080557   0.025912   -3.109 0.001961 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2894 on 640 degrees of freedom
Multiple R-squared:  0.5745,    Adjusted R-squared:  0.5692
F-statistic: 108 on 8 and 640 DF,  p-value: < 2.2e-16
```

Figure_14

Formal Statistical Tests for check the assumptions:

```
===== LOG TRANSFORMATION DIAGNOSTICS =====

1. Shapiro-Wilk Normality Test:
    Shapiro-Wilk normality test
data: residuals(model_log)
W = 0.5814, p-value < 2.2e-16

2. Breusch-Pagan Test (Heteroscedasticity):
    studentized Breusch-Pagan test
data: model_log
BP = 68.091, df = 8, p-value = 1.178e-11

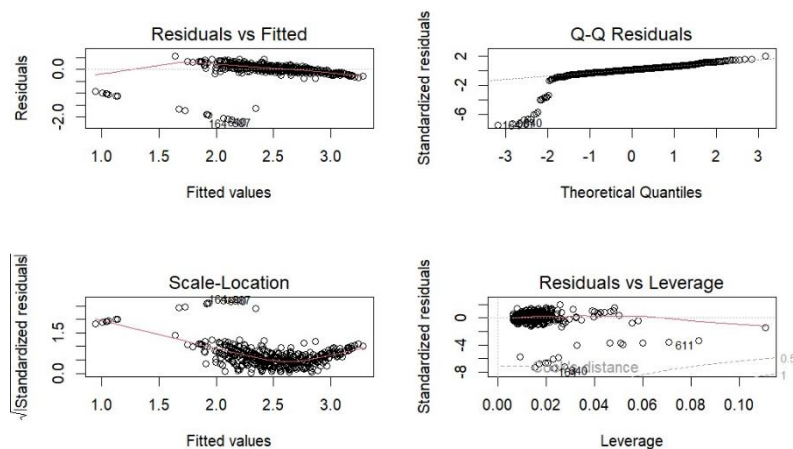
3. VIF (Multicollinearity):
      GVIF Df GVIF^(1/(2*Df))
G2     4.048263 1      2.012030
G1     4.089113 1      2.022156
failures 1.208265 1      1.099211
reason   1.141271 3      1.022268
sex      1.034495 1      1.017101
school   1.181198 1      1.086830

4. Durbin-Watson Test (Autocorrelation):
    Durbin-Watson test
data: model_log
DW = 1.8723, p-value = 0.04371
alternative hypothesis: true autocorrelation is greater than 0

5. Model Fit:
R-squared: 0.5745
Adjusted R-squared: 0.5692
AIC: 243.08
BIC: 287.83
Residual SE: 0.2894
```

Table_10

Residual Plots for the fitted model:



Figure_15

5.1.2.4 Box-Cox Transformation

Fitted Model:

```
Call:
lm(formula = G3_bc ~ G2 + G1 + failures + reason + sex + school,
    data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-31.2792  -3.0396  -0.0446   2.7311  28.4859

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -16.389766    1.137878  -14.404 < 2e-16 ***
G2             3.795624    0.140551   27.005 < 2e-16 ***
G1             1.237157    0.149922    8.252 8.9e-16 ***
failures      -0.424739    0.377128   -1.126  0.2605
reasonhome    -0.170097    0.534948   -0.318  0.7506
reasonother   -1.511192    0.693019   -2.181  0.0296 *
reasonreputation -0.072213    0.547746   -0.132  0.8952
sexM          -0.945417    0.420600   -2.248  0.0249 *
schoolMS      0.007833    0.463961    0.017  0.9865
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.181 on 640 degrees of freedom
Multiple R-squared:  0.8853,    Adjusted R-squared:  0.8838
F-statistic: 617.3 on 8 and 640 DF,  p-value: < 2.2e-16
```

Figure_16

Formal Statistical Tests for check the assumptions:

AIC: 3987.95
BIC: 4032.71
Residual SE: 5.1811
1. Shapiro-Wilk Normality Test:

Shapiro-Wilk normality test

data: residuals(model_bc)
W = 0.93285, p-value < 2.2e-16

2. Breusch-Pagan Test (Heteroscedasticity):

studentized Breusch-Pagan test

data: model_bc
BP = 14.573, df = 8, p-value = 0.06801

3. VIF (Multicollinearity):

	GVIF	Df	GVIF^(1/(2*Df))
G2	4.048263	1	2.012030
G1	4.089113	1	2.022156
failures	1.208265	1	1.099211
reason	1.141271	3	1.022268
sex	1.034495	1	1.017101
school	1.181198	1	1.086830

4. Durbin-Watson Test (Autocorrelation):

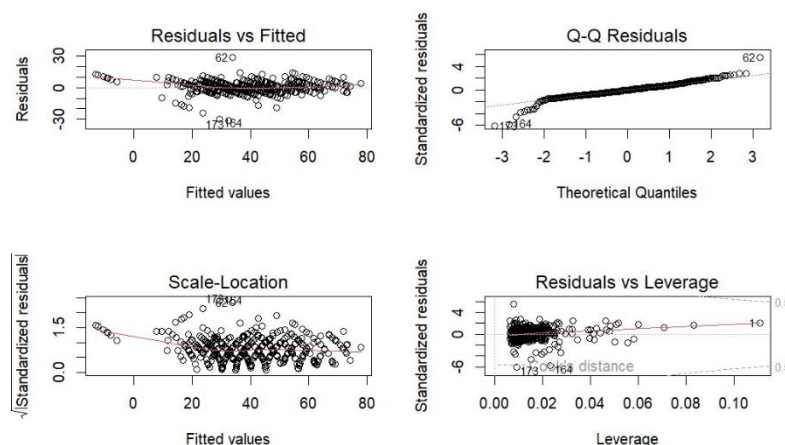
Durbin-Watson test

data: model_bc
DW = 1.7939, p-value = 0.003372
alternative hypothesis: true autocorrelation is greater than 0

5. Model Fit:
R-squared: 0.8853
Adjusted R-squared: 0.8838
AIC: 3987.95
BIC: 4032.71
Residual SE: 5.1811

Table_11

Residual Plots for the fitted model:



Figure_17

5.2 Interpret Findings

5.2.1 Original fitted model – diagnostic results and assumption violations.

The diagnostic results indicate that the assumptions of the fitted linear regression model are not adequately satisfied.

- The Shapiro–Wilk test was significant ($p < 2.2 \times 10^{-16}$), indicating that the residuals are not normally distributed.
- The Breusch–Pagan test was also significant ($p = 0.000414$), providing evidence of heteroscedasticity.
- Furthermore, the Durbin–Watson test ($DW = 1.8628$, $p = 0.03315$) indicates evidence of positive autocorrelation based on the one-sided alternative used in the test.
- The diagnostic plots also support the presence of assumption violations, particularly non-normality and non-constant variance.

5.2.2 Square-root transformation:

The square-root transformation produced a statistically significant overall regression model ($F = 242.2$, $p < 2.2 \times 10^{-16}$) and explained approximately 75.17% of the variation in the transformed response variable ($R^2 = 0.7517$; adjusted $R^2 = 0.7486$).

Among the predictors, G2, failures, reason = other, and school = MS were statistically significant at the 5% level, while G1, reason = home, reason = reputation, and sex were not significant.

However, the diagnostic assessment showed that the square-root transformation did not adequately satisfy the regression assumptions.

- The Shapiro–Wilk test was significant ($W = 0.64667$, $p < 2.2 \times 10^{-16}$), indicating non-normality of the residuals.
- The Breusch–Pagan test was also significant ($BP = 44.698$, $p = 4.199 \times 10^{-7}$), providing evidence of heteroscedasticity.
- In contrast, the Durbin–Watson test was not significant ($DW = 1.8874$, $p = 0.06485$), indicating no significant evidence of positive autocorrelation.
- The VIF values were relatively low, suggesting that multicollinearity was not a serious concern.

The diagnostic plots further support the violations of normality and constant variance, particularly through the substantial departure from the reference line in the Q–Q plot and the changing residual spread in the Scale–Location plot.

Therefore, despite the relatively high R^2 , the square-root transformed model was not considered adequate, and further transformations were investigated.

5.2.3 Cube-Root Transformation:

The cube-root transformed regression model was statistically significant overall ($F = 185.4$, $p < 2.2 \times 10^{-16}$) and explained approximately 69.85% of the variation in the transformed response variable ($R^2 = 0.6985$; adjusted $R^2 = 0.6948$).

The significant predictors at the 5% level were G2, G1, failures, reason = other, and school = MS, while the remaining predictors were not statistically significant.

However, the cube-root transformation did not adequately satisfy the regression assumptions.

- The Shapiro–Wilk test was significant ($W = 0.61779$, $p < 2.2 \times 10^{-16}$), indicating non-normal residuals.
- Similarly, the Breusch–Pagan test was significant ($BP = 52.674$, $p = 1.247 \times 10^{-8}$), indicating heteroscedasticity.
- The Durbin–Watson test was not significant at the 5% level ($DW = 1.8836$, $p = 0.05889$), indicating no significant evidence of positive autocorrelation.
- VIF values also indicated that multicollinearity was not a serious concern.

The diagnostic plots support these findings, particularly the substantial deviation in the Q–Q plot and the non-constant residual spread in the Scale–Location plot.

Therefore, the cube-root transformation did not resolve the major assumption violations, and another transformation was considered.

5.2.4 Logarithmic Transformation:

The logarithmic transformed regression model was statistically significant overall ($F = 108.0$, $p < 2.2 \times 10^{-16}$) and explained approximately 57.45% of the variation in the transformed response variable ($R^2 = 0.5745$; adjusted $R^2 = 0.5692$).

The significant predictors at the 5% level were G2, G1, failures, reason = other, and school = MS, while the remaining predictors were not statistically significant.

However, the logarithmic transformation did not adequately satisfy the regression assumptions.

- The Shapiro–Wilk test was significant ($W = 0.5814$, $p < 2.2 \times 10^{-16}$), indicating non-normal residuals.
- The Breusch–Pagan test was also significant ($BP = 68.091$, $p = 1.178 \times 10^{-11}$), indicating heteroscedasticity.
- In addition, the Durbin–Watson test was significant ($DW = 1.8723$, $p = 0.04371$), providing evidence of positive autocorrelation.
- VIF values indicated that multicollinearity was not a serious concern.

The diagnostic plots support these findings, showing substantial departure from normality in the Q–Q plot and evidence of non-constant residual variance in the Residuals vs Fitted and Scale–Location plots.

Therefore, the logarithmic transformation did not resolve the major regression assumption violations and was not considered an adequate model.

5.2.5 Box-Cox Transformation:

The Box–Cox transformed regression model was statistically significant overall ($F = 617.3$, $p < 2.2 \times 10^{-16}$) and explained approximately 88.53% of the variation in the transformed response variable ($R^2 = 0.8853$; adjusted $R^2 = 0.8838$).

The significant predictors at the 5% level were G2, G1, reason = other, and sex = M, while failures, reason = home, reason = reputation, and school = MS were not statistically significant.

The diagnostic assessment showed mixed results.

- The Shapiro–Wilk test was significant ($W = 0.93285$, $p < 2.2 \times 10^{-16}$), indicating that the residuals were still not normally distributed.
- However, the Breusch–Pagan test was not significant ($BP = 14.573$, $p = 0.06801$), indicating no significant evidence of heteroscedasticity and suggesting that the homoscedasticity assumption was satisfied.
- The Durbin–Watson test was significant ($DW = 1.7939$, $p = 0.003372$), indicating evidence of positive autocorrelation.
- VIF values remained relatively low, suggesting that multicollinearity was not a serious concern.

The diagnostic plots also show departures from normality, particularly in the Q–Q plot, although the variance pattern appears improved compared with the other transformations.

Therefore, the Box–Cox transformation improved the homoscedasticity of the model but did not resolve all the assumption violations, particularly normality and independence.

Consequently, the Box–Cox transformed model was not considered fully adequate as the final linear regression model.

5.2.6 Overall Interpretation of the Transformation Results

Overall, none of the transformations fully satisfied all the assumptions required for the linear regression model.

The square-root and cube-root transformations produced statistically significant models with R^2 values of 0.7517 and 0.6985, respectively. However, both models continued to show significant violations of normality and homoscedasticity, although there was no significant evidence of positive autocorrelation.

The logarithmic transformation performed comparatively poorly, with a lower R^2 of 0.5745, and violations of normality, homoscedasticity, and independence remained evident.

Among the transformations considered, the Box–Cox transformation provided the highest explanatory power, with an R^2 of 0.8853 and adjusted R^2 of 0.8838. It also improved the constant variance assumption, as the Breusch–Pagan test was not statistically significant ($p =$

0.06801). However, the Shapiro–Wilk test remained significant ($p < 2.2 \times 10^{-16}$), indicating non-normality of the residuals, while the Durbin–Watson test ($p = 0.003372$) indicated evidence of positive autocorrelation.

Therefore, despite applying square-root, cube-root, logarithmic, and Box–Cox transformations, the key assumptions of multiple linear regression were not adequately satisfied. This indicates that MLR is not an appropriate modelling approach for these data. Consequently, an alternative modelling method more suitable for the characteristics of the response variable was required.

5.3 Compare Results with Previous Studies

The findings of the present study can be compared with previous research, examining factors associated with students' academic performance.

Zubair et al. (2024) found that study time was associated with academic performance, while Gong et al. (2025) reported an association between high-frequency social-media use and poorer academic performance.

Similarly, Faught et al. (2017) identified relationships between lifestyle behaviors and academic performance, and Riaz et al. (2023) reported associations between social-media and entertainment-related behaviors and academic success.

These studies mainly focused on specific academic, behavioral, social, or lifestyle factors, whereas the present study considered a broader combination of academic, demographic, family, social, and school-related characteristics simultaneously.

The present findings are particularly noteworthy when compared with Oughdir et al. (2020), who successfully applied multiple linear regression with k-fold cross-validation to predict student performance.

In contrast, the present study found that the assumptions required for multiple linear regression were not adequately satisfied. Although square-root, cube-root, logarithmic, and Box–Cox transformations were applied, none successfully resolved all the identified assumption violations.

Therefore, a practically suitable multiple linear regression model could not be established for the present dataset. This comparison highlights that, although MLR has been useful in previous studies of student performance, its suitability depends on the characteristics of the specific dataset and the extent to which its underlying assumptions are satisfied.

5.4 Discuss Implications of the Findings

The analysis showed that previous grades (G1 and G2) and previous failures repeatedly appeared as potentially important predictors of final academic performance (G3). This suggests that students' previous academic performance may provide useful information for understanding their final outcomes. The reason for choosing the school, particularly the

“other” category, also appeared relevant in several models, although its practical meaning is difficult to determine because the category may represent different underlying reasons.

However, these findings should be interpreted cautiously because a practically suitable multiple linear regression model could not be established. Although some fitted models had relatively high R^2 values and statistically significant predictors, the required MLR assumptions were not adequately satisfied. This highlights that model suitability should not be judged solely by explanatory power or statistical significance; diagnostic results and model assumptions must also be considered before drawing conclusions or making predictions.

Limitations:

The main limitation of this study is that important MLR assumptions remained violated even after applying square-root, cube-root, logarithmic, and Box–Cox transformations. Therefore, the estimated effects and significance of the predictors cannot be considered fully reliable for final inference.

In addition, some predictors showed borderline significance, while the “reasonother” category was difficult to interpret because it combines unspecified reasons for school choice.

Finally, the analysis was limited to MLR and response transformations, and alternative modelling approaches were beyond the scope of the present study.

6. Conclusion and Recommendations

6.1 Summary of Major Findings

- A practically suitable MLR model could not be established.
The original fitted multiple linear regression model violated important regression assumptions.
- Response transformations did not fully solve the assumption violations.
Square-root, cube-root, logarithmic, and Box–Cox transformations were investigated, but none satisfied all the required MLR assumptions.
- The Box–Cox transformation showed the greatest improvement.
It produced the highest R^2 (0.8853) and resolved the heteroscedasticity issue, but normality and independence problems remained. Therefore, it still could not be accepted as an adequate final MLR model.
- Previous academic performance consistently appeared important.
G1 and/or G2 repeatedly appeared as important predictors across the fitted models, suggesting a strong association between previous grades and final grade (G3).
- Previous failures also appeared potentially important.
The variable failures were significant in several fitted models, suggesting that students’ previous failure history may be associated with their final academic performance.

- Multicollinearity was not a major problem.
The VIF results indicated that serious multicollinearity was not present in the fitted models.
- High explanatory power alone did not indicate an adequate model.
Some models achieved relatively high R^2 values, but diagnostic testing still showed assumption violations. Therefore, model adequacy could not be determined solely from R^2 or statistical significance.

6.2 Recommendations

Based on the findings of this study, it is recommended that the fitted multiple linear regression models should not be used as final predictive models because important regression assumptions were not adequately satisfied. Future analyses should consider alternative modelling approaches that may be more suitable for the characteristics of the final grade (G3). Greater attention should also be given to previous academic performance, particularly G1, G2, and previous failures, as these variables repeatedly appeared important during the modelling process. In addition, broad categorical variables such as the “other” category under reason for school choice should be defined more clearly to improve their practical interpretation. Finally, future studies should conduct thorough model diagnostics before accepting a model and, where possible, use larger and more diverse student samples to improve the generalizability of the findings.

6.3 Suggested Areas for Future Research

Future research should explore alternative modelling approaches that may be more appropriate for the characteristics of students’ final grade (G3), since the assumptions of multiple linear regression could not be adequately satisfied in the present study.

Further studies could also investigate the influence of previous academic performance, particularly G1, G2, and previous failures, as these variables repeatedly appeared important during the modelling process.

In addition, future research could include more clearly defined explanatory variables, particularly for broad categories such as the reason for school choice and use larger and more diverse student populations to determine whether the findings are consistent across different educational settings.

7. References

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Dataset Reference

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